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CODE A

Mock JEE Main – 8 (CBT) | JEE - 2024

Date: 18/03/2024 **Maximum Marks: 300**

Timing: 3:30 PM to 6:30 PM **Duration : 3.0 Hours**

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300.**
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices **(A)**, **(B)**, **(C)** and **(D)**, out of which **ONLY ONE CHOICE** is correct.
4. **Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted. You *will* NOT be allowed to attempt the sixth question. If you wish to attempt any other question apart from the five already attempted, then you will have to delete any one response from the five previously answered and then proceed to answer the new one.**
   The answer to each question should be **rounded off to the nearest integer**.
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them**.

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.

Name of the Candidate (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

PART I: PHYSICS	MARKS: 100

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Visible light of wavelength 6000×10^{-8} cm falls normally on a single slit and produces a diffraction pattern. It is found that the second diffraction minimum is at 60° from the central maximum. If the first minimum is produced at θ_1 then θ_1 is close to

(A) 25° **(B)** 30° **(C)** 10° **(D)** 45°

2. A plane electromagnetic wave travelling along the *x*-direction has a wavelength of 3 mm. The variation in the electric field occurs in the *y*-direction with an amplitude $66Vm^{-1}$. The equations for the electric and magnetic fields as a function of *x* and *t* respectively are:

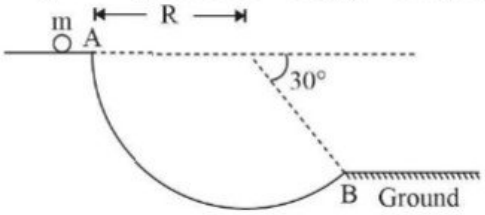
(A) $E_y = 66\cos\pi\times10^{11}\left(t-\frac{x}{c}\right),\ B_z = 1.1\times10^{-7}\cos\pi\times10^{11}\left(t-\frac{x}{c}\right)$

(B) $E_y = 11\cos2\pi\times10^{11}\left(t-\frac{x}{c}\right),\ B_z = 11\times10^{-7}\cos2\pi\times10^{11}\left(t-\frac{x}{c}\right)$

(C) $E_x = 33\cos\pi\times10^{11}\left(t-\frac{x}{c}\right),\ B_z = 11\times10^{-7}\cos\pi\times10^{11}\left(t-\frac{x}{c}\right)$

(D) $E_y = 66\cos2\pi\times10^{11}\left(t-\frac{x}{c}\right),\ B_z = 2.2\times10^{-7}\cos2\pi\times10^{11}\left(t-\frac{x}{c}\right)$

3. A small particle is gently pushed from point A. It goes through a smooth spherical surface. Find the maximum height attained from ground. [Given : R = 1m, g = 10 m/s^2].

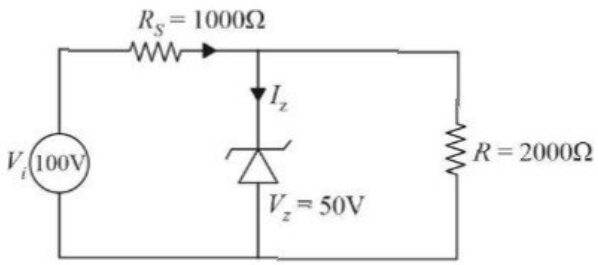


(A) $\frac{1}{2}m$ **(B)** $\frac{3}{8}m$ **(C)** $1m$ **(D)** $2m$

4. In a series LR circuit with *AC* source, the voltage drop across inductor is 8 volt and across resistor is 6 volt. Then voltage applied and power factor of circuit respectively are :

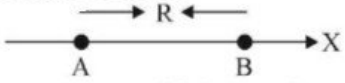
(A) 14V, 0.8 **(B)** 10V, 0.8 **(C)** 10V, 0.6 **(D)** 14V, 0.6

5. For the circuit shown below, calculate the value of I_z.



(A) 25 mA **(B)** 0.15 A **(C)** 0.1 A **(D)** 0.05 A

6. Two electric dipoles, *A*, *B* with respective dipole moments $d_A = -4qa\hat{i}$ & $d_B = -2qa\hat{i}$ are placed on the *x*-axis with a separation *R*, as shown in the figure



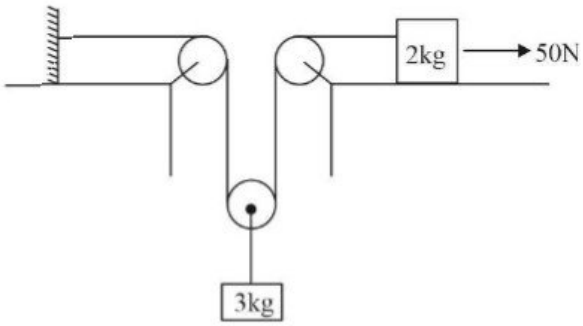
The distance from A on *x*-axis where net potential is zero, is:

(A) $\frac{\sqrt{2}R}{\sqrt{2}+1}$ **(B)** $\frac{\sqrt{2}R}{\sqrt{2}-1}$ **(C)** $\frac{R}{\sqrt{2}+1}$ **(D)** $\frac{R}{\sqrt{2}-1}$

7. Find bulk modulus of elasticity of 'He' gas in adiabatic condition if pressure of gas is *P*.

(A) $\left(\frac{7}{5}P\right)$ **(B)** $\left(\frac{1}{3}P\right)$ **(C)** $\left(\frac{5}{3}P\right)$ **(D)** $2P$

8. In the figure shown, if all surfaces are smooth, pulleys and strings are ideal, find acceleration of 3kg block.



(A) $\frac{70}{11} m/s^2$

(B) $\frac{35}{11} m/s^2$

(C) $\frac{9}{11} m/s^2$

(D) $\frac{10}{11} m/s^2$

9. A particle is projected with speed u at an angle θ with horizontal. Find the magnitude of angular momentum of particle about point of projection at the time just before it hits the ground.

(A) $\frac{mu^3}{g}\sin\theta\cos^2\theta$ **(B)** $\frac{mu^2}{2g}\sin\theta\cos^2\theta$

(C) $\frac{2mu^3}{g}\sin^2\theta\cos\theta$ **(D)** $\frac{mu^3}{g}\sin^2\theta\cos\theta$

10. The molar heat capacity of a monoatomic ideal gas following below mentioned process is:

[$TV^{3/2}$ = constant, R = Universal gas constant]

(A) $\frac{7}{5}R$ **(B)** $\frac{5}{6}R$ **(C)** $\frac{3}{2}R$ **(D)** $\frac{2}{3}R$

11. The rms speed of a gas containing 100 molecules such that speed of ith molecule is $2i$ m/s.

(A) $\sqrt{13534}\, m/s$ **(B)** $\sqrt{13435}\, m/s$ **(C)** 100 m/s **(D)** $\sqrt{12121}\, m/s$

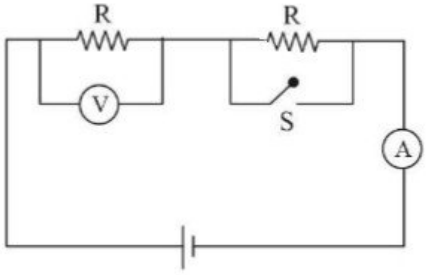
12. Match the column I and II, and choose the correct combination from the option given:

Column I		Column II	
I.	Young's modulus	**P.**	$[M^{-1}L^{-3}T^4A^2]$
II.	Coefficient of viscosity	**Q.**	$[M^0LT^{-1}]$
III.	Permittivity of free space	**R.**	$ML^{-1}T^{-1}$
IV.	Terminal velocity	**S.**	$ML^{-1}T^{-2}$

(A) I – Q ; II – R ; III – S ; IV – P **(B)** I – S ; II – R ; III – P ; IV – Q

(C) I – P ; II – Q ; III – R ; IV – S **(D)** I – R ; II – S ; III – Q ; IV – P

13. In the circuit as shown in figure, V is high resistance voltmeter and A is a low resistance ammeter. Switch S is open. What is the effect on the voltmeter and ammeter readings when switch *S* is closed?



(A) Reading of voltmeter and ammeter increases

(B) Voltmeter reading increases and ammeter reading decreases

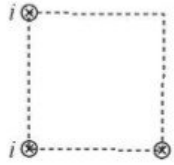
(C) Voltmeter reading increases while the ammeter reading remains same

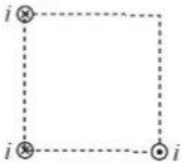
(D) Voltmeter reading decreases and ammeter reading decreases

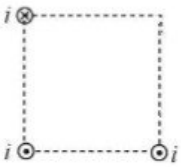
14. The secondary coil of an ideal step down transformer is delivering 500 watt power at 12.5 A current. If the ratio of turns in the primary to the secondary is 5 : 1, then the current flowing in the primary coil will be :

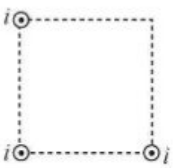
(A) 62.5 A **(B)** 2.5 A **(C)** 6 A **(D)** 0.4 A

15. Three long wires, with identical currents either directly into or directly out of the page, are placed at three corners of a square in four different arrangements as shown. Correct order of the magnitude of net magnetic field at the empty upper right corner of the square is :

I. 

II. 

III. 

IV. 

(A) $B_I = B_{IV} > B_{II} = B_{III}$ **(B)** $B_I > B_{IV} > B_{II} = B_{III}$

(C) $B_{II} = B_{III} > B_I = B_{IV}$ **(D)** $B_I > B_{III} > B_{II} > B_{IV}$

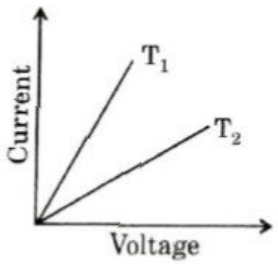
16. The maximum kinetic energy of photoelectrons emitted from a surface when photons of energy of 6eV fall on it is 4eV. The stopping potential (in volts) is :

(A) 2 **(B)** 4 **(C)** 6 **(D)** 10

17. Gravitational potential and acceleration due to gravity is V and g at a point in gravitational field of a spherical planet of radius R. The height of this point from surface is :

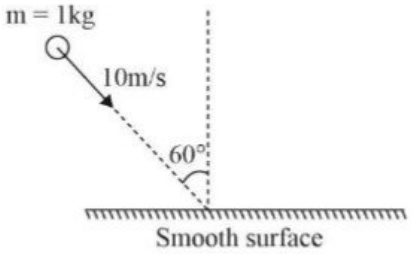
(A) $\left(\frac{V}{g} - R\right)$ **(B)** $\left(\frac{g}{V} - R\right)$ **(C)** $\left(\frac{V}{gR}\right)$ **(D)** $(gR - V)$

18. The current in a metallic conductor is plotted against voltage at two different temperatures T_1 & T_2. Which one of the following is correct?



(A) $T_1 > T_2$ **(B)** $T_1 < T_2$ **(C)** $T_1 = T_2$ **(D)** None of the above

19. A ball of mass 1 kg hits a smooth surface as shown in figure. Collision is inelastic with $e = \frac{1}{2}$. Find impulse imparted by surface on ball.



(A) 10 N-s **(B)** 7.5 N-s **(C)** 5 N-s **(D)** 0 N-s

20. In a hydrogen atom, the binding energy of the electron in n^{th} state is E_n, then the frequency of revolution of the electron in the nth orbit is :

(A) $\frac{2E_n}{nh}$ **(B)** $\frac{2E_n n}{h}$ **(C)** $\frac{E_n}{nh}$ **(D)** $\frac{E_n n}{h}$

SPACE FOR ROUGH WORK

SECTION-2

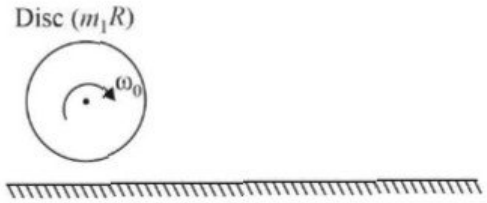
Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.
The answer to each question should be **rounded off to the nearest integer**.

1. A $4\mu F$ capacitor is charged by a 200 V supply. It is then disconnected from the supply and is connected to another uncharged $2\mu F$ capacitor. How much electrostatic energy of the first capacitor is lost in the form of heat and electromagnetic radiation? (in mJ)

2. A particle moving along x-axis, is situated at $x = 2m$ at $t = 0$; and moving towards (+ve) x-axis with speed of 9 m/s. Acceleration of the particle is $2m/s^2$ directed along negative x-axis. The distance travelled by the particle in 5th sec is $x\,m$. Find the value of $2x$.

3. In a hydrogen atom, the electron is in n^{th} excited state. It may come down to second excited state by emitting ten different wavelengths. What is the value of n?

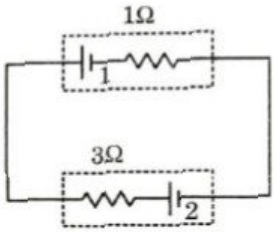
4. A disc of mass m and radius R is rotated with angular velocity ω_0, and kept gently on rough horizontal surface. After some time it starts pure rolling with angular velocity ω. Find $\left(\frac{\omega_0}{\omega}\right)$.



5. In a closed organ pipe, the frequency of fundamental mode is 40 Hz. A certain amount of water is now poured in the organ pipe so that the fundamental frequency is increased to 60 Hz. If the organ pipe has a cross-sectional area of 16 cm^2, the amount of water poured in the organ pipe is x kg. Find the value of $10x$. (Assume speed of sound as 330 m/s)

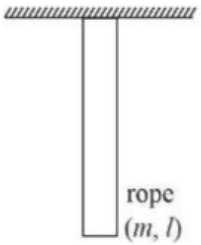
6. A circular coil of radius 10 cm is placed in a uniform magnetic field of $3.0\times10^{-5}T$ with its plane perpendicular to the field initially. It is rotated at a constant angular speed about an axis along the diameter of coil and perpendicular to magnetic field, so that it undergoes half of rotation in 0.2s. The maximum value of emf induced (in μV) in the coil will be close to the integer ___. (Take : $\pi^2 = 10$)

7. In the figure shown, battery 1 has e.m.f = 6V and internal resistance $= 1\Omega$. Battery 2 has e.m.f. = 2V and internal resistance $= 3\Omega$. The wires have negligible resistance. What is the potential difference (in V) across the terminals of battery 2?



8. The same size images are formed by a convex lens when the object is placed at 20 cm or at 10 cm from the lens. The focal length of convex lens is _____cm.

9. A uniform rope of mass m and length l is hanging from celling. Elastic potential energy stored in rope is $\left(\frac{m^2g^2l}{nYA}\right)$, where Y is Young's modulus of elasticity and A is cross sectional area. Find value of n.



10. A coil in the shape of an equilateral triangle of side 10 cm lies in a vertical plane between the pole pieces of permanent magnet producing a horizontal magnetic field 20 mT. The torque acting on the coil when a current of 0.2 A is passed through it and its plane becomes parallel to the magnetic field will be $\sqrt{x}\times10^{-5}$ Nm. The value of x is _____.

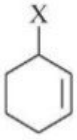
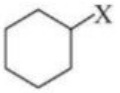
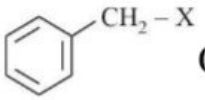
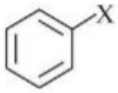
SPACE FOR ROUGH WORK

PART II: CHEMISTRY	MARKS: 100

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. From which of the following replacement of X^- (halide ion) is most difficult.

(A)  **(B)**  **(C)**  **(D)** 

2. Etard's reaction in the following is :

(A) Dry distillation of calcium benzoate

(B) Reaction of benzene with Cl_2 in the presence of UV light

(C) Oxidation of toluene by chromyl chloride followed by hydrolysis

(D) Oxidation of toluene by alkaline $KMnO_4$

3. Which of the following compounds show optical isomerism?

I. $cis-[Co(NH_3)_4Cl_2]^+$ **II.** $trans-[Co(en)_2Cl_2]^+$

III. $cis-[Co(en)_2Cl_2]^+$ **IV.** $[Co(en)_3]^{3+}$

Select the correct answer using the codes given below:

(A) I and II **(B)** II and III **(C)** III and IV **(D)** I, III and IV

4. $$\text{Phthalimide} \xrightarrow[\text{(ii) } CH_3CH_2Cl \\ \text{(iii) } NaOH(aq.)]{\text{(i) KOH}} X \xrightarrow[KOH,\ \Delta]{CHCl_3} Y$$

[Y] as the major product is

(A) $CH_3-CH_2-NH_2$ **(B)** $Ph-CH_2-NC$

(C) CH_3-CH_2-CN **(D)** CH_3-CH_2-NC

5. Statement 1 : Sulphate is estimated as $BaSO_4$ and not as $MgSO_4$

Statement 2 : Ionic radius of Mg^{2+} is smaller than that of the Ba^{2+}

(A) Statement 1 is true and Statement 2 is false

(B) Statement 1 is false and statement 2 is true

(C) Both statements are true and statement 2 is correct explanation of statement 1

(D) Both statements are true and statement 2 is incorrect explanation of statement 1

6. Which of the following group of ions is paramagnetic in nature :

(A) Cu^+, Zn^{2+}, Sc^{3+} **(B)** $Mn^{2+}, Fe^{3+}, Ni^{2+}$

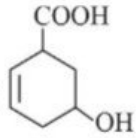
(C) $Cr^{2+}, Mn^{3+}, Sc^{3+}$ **(D)** $Cu^{2+}, Ni^{2+}, Ti^{4+}$

7. Match the column I and II, and choose the correct combination from the option given

Column I		Column II	
I.	NH_2^-	**P.**	Square pyramidal
II.	$XeOF_2$	**Q.**	V-shaped
III.	ICl_4^-	**R.**	T-shaped
IV.	$[SbF_5]^{2-}$	**S.**	Square planar

(A) I – P; II – Q; III – R; IV – S **(B)** I – Q; II – R; III – S; IV – P

(C) I – R; II – P; III – Q; IV – S **(D)** I – P; II – S; III – R; IV – Q

8. What is correct IUPAC name of 

(A) 3-hydroxy cyclohex-5-ene-1-carboxylic acid

(B) 3-hydroxy cyclohex-5-ene-1-oic acid

(C) 5-hydroxy cyclohex-2-ene-1-carboxylic acid

(D) 5-hydroxy cyclohex-2-ene-1-oic acid

9. Statement 1 : The first ionization energy of Be is greater than that of B.
Statement 2 : 2p orbital is lower in energy than 2s.

(A) Statement 1 is true and Statement 2 is false

(B) Statement 1 is false and statement 2 is true

(C) Both statements are true and statement 2 is correct explanation of statement 1

(D) Both statements are true and statement 2 is incorrect explanation of statement 1

10. **Statement-1:** Alcoholic KOH converts CH_3-CH_2Br into ethene while aqueous KOH converts CH_3-CH_2-Br into ethyl alcohol.

Statement-2: Alcoholic KOH is more basic than aqueous KOH.

(A) Statement 1 is true and Statement 2 is false

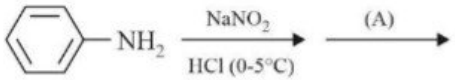
(B) Statement 1 is false and statement 2 is true

(C) Both statements are true and statement 2 is correct explanation of statement 1

(D) Both statements are true and statement 2 is incorrect explanation of statement 1

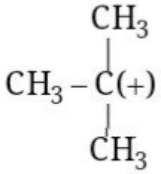
11. Out of the following species which doesn't have fully filled f subshell.

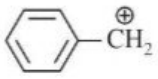
(A) Yb **(B)** Eu **(C)** No **(D)** Lr

12.  Aniline yellow. A will be :

(A) Phenol **(B)** β -Napthol **(C)** α -Napthol **(D)** Aniline

13. Which of the following carbocation is most stable?

(A) 

(B) 

(C) 

(D) $CH_3-\overset{\oplus}{C}H-OCH_3$

14. Major product in following reaction is :

$CH_3CH=CH_2 \xrightarrow[2.\ H_2O_2/OH^-]{1.\ B_2D_6}$

(A) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_2D$

(B) $CH_3-\underset{\underset{D}{|}}{CH}-CH_2OH$

(C) $CH_3-\underset{\underset{OD}{|}}{CH}-CH_3$

(D) $CH_3-CH_2-CH_2-OD$

15. Given below two statements :

Statement-1 : For He^+ ion, the energy of 3p orbital is greater than that of 3s orbital.

Statement-2 : Energy of He^+ orbitals can be assessed using $(n+\ell)$ rule.

(A) Statement 1 and Statement 2 are false
(B) Statement 1 and statement 2 are true
(C) Both statements are true and statement 2 is correct explanation of statement 1
(D) Both statements are true and statement 2 is incorrect explanation of statement 1

16. During depression of freezing point in a solution, the following are in equilibrium :

(A) Liquid solvent – solid solvent **(B)** Liquid solvent – solid solute
(C) Liquid solute – solid solute **(D)** Liquid solute – solid solvent

17. Co^{2+} ion forms complex with SCN^- ions having geometry:

(A) Tetrahedral **(B)** Square planar **(C)** Octahedral **(D)** Trigonal bipyramidal

18. Kjeldahl method for estimation of nitrogen is applicable to:

(A) Nitrobenzene **(B)** Benzene diazonium chloride
(C) Pyridine **(D)** Benzylamine

19.

$$Ph-\underset{\underset{O}{\|}}{C}-Cl \xrightarrow{[X]} Ph-\underset{\underset{O}{\|}}{C}-H$$

$$Ph-CN \xrightarrow{[Y]} Ph-\underset{\underset{O}{\|}}{C}-H$$

$$Ph-\underset{\underset{O}{\|}}{C}-OEt \xrightarrow{[Z]} Ph-\underset{\underset{O}{\|}}{C}-H$$

[X], [Y] and [Z] respectively are :

(A) (i) $SnCl_2/HCl$; H_3O^+ ; (i) $DIBA\ell-H$/(ii) H_2O ; $H_2/Pd-BaSO_4$ (ii) H_3O^+

(B) $H_2/Pd-BaSO_4$; (i) $SnCl_2/HCl$; (i) $DIBA\ell-H$/(ii) H_2O (ii) H_3O^+

(C) (i) $DIBA\ell-H$/(ii) H_2O ; (i) $SnCl_2/HCl$; $H_2/Pd-BaSO_4$ (ii) H_3O^+

(D) (i) $DIBA\ell-H$/ (ii) H_2O ; $H_2/Pd-BaSO_4$; (i) $SnCl_2/HCl$ (ii) H_3O^+

20. Which of the following carbohydrates develops blue colour on treatment with iodine solution?

(A) Glucose **(B)** Amylose **(C)** Maltose **(D)** Fructose

SPACE FOR ROUGH WORK

SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.
The answer to each question should be **rounded off to the nearest integer**.

1. How many electrons are present in bonding molecular orbitals of O_2^+ ?

2. A 0.25 M solution of pyridinium chloride $C_5H_5NH^+Cl^-$ was found to have a pH of 2.75. The value of pK_b for pyridine, (C_5H_5N) is x. What is value of x?

3. In the following reaction
$4Zn + yHNO_3(dil) \longrightarrow aZn(NO_3)_2 + bH_2O + cNH_4NO_3$
What is the sum of the coefficients $(a + b + c)$?

4. For the reaction $\frac{1}{2}N_2(g) + \frac{3}{2}H_2(g) \longrightarrow NH_3(g)$; $\Delta H = -30kJ$ equilibrium will be attained at 477°C. If standard entropy of $N_2(g)$ and $NH_3(g)$ are 60 and 50J $mol^{-1}K^{-1}$ respectively then calculate the standard entropy of $H_2(g)$ in $Jmole^{-1}K^{-1}$.

5. How many of the following compounds will not give positive iodoform test?
Acetophenone, Benzophenone, 2-Pentanone, 3-Pentanone, Acetaldehyde, CH_3COCH_3, $(CH_3)_2CHOH$, $(CH_3CH_2)_2CHOH$, CH_3COOH, CH_3CONH_2, CH_3COOCH_3, CH_3COCl

6. In Carius method of estimation of halogen, 0.15 g of an organic compound gave 0.12 g of AgBr. What is the percentage of bromine in the compound? (Round off to nearest integer)
[Given : Atomic mass Ag-108 ; Br-80]

7. If IUPAC name of an element is "ununpentium" then the element belongs to n^{th} group of modern periodic table. The value of n is _____ .

8. For a first order reaction, time required for 99% completion is x times the time required for the completion of 90% of the reaction. Find the value of x.

9. 92 g mixture of $CaCO_3$ and $MgCO_3$ heated strongly in an open vessel. After complete decomposition of the carbonates it was found that the weight of residue left behind is 48 g. Find the mass of $MgCO_3$ (in grams) in the mixture.

10. H_3PO_4 $(98\ g\ mol^{-1})$ is 98% by mass of solution. If the density of solution is 1.8 g/ml, the molarity of solution is _____.

SPACE FOR ROUGH WORK

PART III: MATHEMATICS	MARKS: 100

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. If three unit vectors $\vec{a},\vec{b}$ & $\vec{c}$ are such that $\vec{a}\times\vec{b}=\vec{c}, \vec{b}\times\vec{c}=\vec{a}, \vec{c}\times\vec{a}=\vec{b}$ the answer the following question. If the vectors $3\vec{a}-2\vec{b}+2\vec{c}$ & $-\vec{a}-2\vec{c}$ are adjacent sides of a parallelogram, then the acute angle between the diagonal is :

(A) $\frac{\pi}{4}$ **(B)** $\frac{\pi}{3}$ **(C)** $\frac{\pi}{2}$ **(D)** $\frac{2\pi}{3}$

2. If the quadratic equation $x^2+(2-\tan\theta)x-(1+\tan\theta)=0$ has two integral roots and sum of all possible value of θ in the interval $(0, 2\pi)$ is $k\pi$, then the value of k is equal to _____.

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

3. Given $p,q\in\{0,1,2,3,4,.....,19,20\}$. Consider the system of equations $x+y+z=4$, $2x+y+3z=6$, $x+2y+pz=q$, Then which one is incorrect.

(A) The number of ordered pairs (p, q) so that the system of equations has unique solution is 420
(B) The number of ordered pairs (p, q) so that the system of equations has no solution is 20
(C) The number of ordered pairs (p, q) so that system of equations has infinite solutions is 1
(D) For unique solution minimum value of $p + q = 0$

4. Find the area of the region bounded by the curves $y=x-1$ & $(y-1)^2=4(x+1)$:

(A) $\frac{5}{3}$ **(B)** $\frac{10}{3}$ **(C)** $\frac{11}{3}$ **(D)** $\frac{64}{3}$

5. Median for the following data is:

Height (in cm)	95 – 105	105 – 115	115 – 125	125 – 135	135 – 145	145 – 155
Number of boys	9	13	25	30	13	10

(A) 125 **(B)** 126 **(C)** 127 **(D)** 128

6. A piece of cheese is located at (12,10) in a coordinate plane. A mouse is at $(4,-2)$ and is running up along the line $y=-5x+18$ to get closer to the piece of cheese. At the point (a, b) the mouse starts getting farther from the cheese rather than closer to it. Then the value of $(a+b)$ is :

(A) 6 **(B)** 10 **(C)** 18 **(D)** 14

7. Find the domain of the function where [.] denotes greatest integer function.

$f(x)=\log_{(4-x)}(x-1)-\sin^{-1}[2x-3]$ is :

(A) (1, 2) **(B)** (1, 2.5) **(C)** (1, 1.5) **(D)** (3, 4)

8. If $a_1,a_2,.....,a_n$ is a sequence of positive numbers which are in A.P. with common difference d and $a_1+a_4+a_7+.....+a_{16}=147$. Maximum value of $a_1a_2.....a_{16}$ is $\left(\frac{S}{W}\right)^{16}$ (where S and W are coprime), then $S+W$ is :

(A) 100 **(B)** 98 **(C)** 49 **(D)** 51

9. Find $a+b$ if $\lim\limits_{x\to 0}\dfrac{\int_0^x \frac{t^2dt}{\sqrt{a+t}}}{bx-\sin x}=1$:

(A) 3 **(B)** 4 **(C)** 5 **(D)** 2

10. If $f(x)$ is an odd continuous function in $[-2,2]$ and differentiable in $(-2,2)$ then :

(A) $f'(c)$ must be zero for at least one $c\in(-2,0)$

(B) $f'(2)+f'(-2)=0$

(C) $f'(c)$ must be zero for at least one $c\in(0,2)$

(D) $f'(c)=\dfrac{1}{2}f(2)$, for some $c\in(-2,0)$

11. The solution of $x^2\dfrac{dy}{dx}-xy=1+\cos\dfrac{y}{x}$ is $\tan\left(\dfrac{y}{px}\right)=c-\dfrac{1}{qx^2}$ where c is arbitrary constant then, find the value of $p+q$:

(A) 2 **(B)** –3 **(C)** 4 **(D)** –1

12. Let $f(x)=\begin{vmatrix}\sin^2 x & -2+\cos^2 x & \cos 2x\\ 2+\sin^2 x & \cos^2 x & \cos 2x\\ \sin^2 x & \cos^2 x & 1+\cos 2x\end{vmatrix}$, $x\in[0,\pi]$. Then the maximum value of $f(x)$ is equal to :

(A) 3 **(B)** 4 **(C)** 5 **(D)** 6

13. If $PS'P'$ is a focal chord of $\dfrac{x^2}{25}+\dfrac{y^2}{9}=1$ and SP = 3, then the value of $61S'P'-56$, where S and S' are foci of ellipse.

(A) 5 **(B)** 6 **(C)** 7 **(D)** 8

14. Three numbers are chosen at random without replacement from {1, 2, 3, …. , 8}. The probability that their minimum is 3, given that their maximum is 6, is:

(A) $\dfrac{3}{8}$ **(B)** $\dfrac{1}{5}$ **(C)** $\dfrac{1}{4}$ **(D)** $\dfrac{2}{5}$

15. A parallelogram is constructed on $3\vec{a}+\vec{b}$ & $\vec{a}-4\vec{b}$, where $|\vec{a}|=6$ & $|\vec{b}|=8$, and a & $\vec{b}$ are perpendicular. Then the length of the longer diagonal is :

(A) $36\sqrt{2}$ **(B)** $4\sqrt{109}$ **(C)** $24\sqrt{2}$ **(D)** None of these

16. If $|z-2i|\le\sqrt{2}$, where $i=\sqrt{-1}$, then the maximum value of $|3+i(z-1)|$, is :

(A) $\sqrt{2}$ **(B)** $2\sqrt{2}$ **(C)** $2+\sqrt{2}$ **(D)** $3+2\sqrt{2}$

17. If the direction cosines of two lines are such that $l+m+n=0$, $l^2+m^2-n^2=0$, then the angle between them is

(A) π **(B)** $\dfrac{\pi}{3}$ **(C)** $\dfrac{\pi}{4}$ **(D)** $\dfrac{\pi}{6}$

18. Let C_1 & C_2 are circles defined by $x^2+y^2-20x+64=0$ and $x^2+y^2+30x+144=0$. The length of the shortest line segment PQ that is tangent to C_1 at P and C_2 at Q is :

(A) 15 **(B)** 18 **(C)** 20 **(D)** 24

19. Let a be a positive constant number. Consider two curves $C_1: y=e^x, C_2: y=e^{a-x}$. Let S be the area of the part surrounding by C_1, C_2 and the Y-axis, then $\lim\limits_{a\to 0}\dfrac{S}{a^2}$ equals.

(A) 4 **(B)** 1/2 **(C)** 0 **(D)** 1/4

20. If $a,b,c\in R^+$ then $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\dfrac{n}{(k+an)(k+bn)}$ is equal to:

(A) $\dfrac{1}{a-b}\ln\dfrac{b(b+1)}{a(a+1)}$, if $a\neq b$ **(B)** $\dfrac{1}{a-b}\ln\dfrac{a(b+1)}{b(a+1)}$, if $a\neq b$

(C) Non-existent, if $a=b$ **(D)** $\dfrac{1}{a(2+a)}$, if $a=b$

SPACE FOR ROUGH WORK

SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.
The answer to each question should be **rounded off to the nearest integer**.

1. If a variable line has its intercepts on the coordinate axes are e and e', where $\frac{e}{2}$ & $\frac{e'}{2}$ are the eccentricities of a hyperbola and its conjugate hyperbola, then the line always touches the circle $x^2+y^2=r^2$, where r is _____.

2. If the shortest distance between the lines $\frac{x-3}{3}=\frac{y-8}{-1}=\frac{z-3}{1}$ & $\frac{x+3}{-3}=\frac{y+7}{2}=\frac{z-6}{4}$ is $\lambda\sqrt{30}$ unit, then the value of λ is _____.

3. In a town of 10000 families, it was found that 40% families buy newspaper *A*, 20% families buy newspaper *B* and 10% families buy newspaper *C*, 5% families buy newspapers *A* and *B*, 3% buy newspapers *B* and *C* and 4% buy newspapers *A* and *C*. If 2% families buy all three newspapers, then number of families which buy *A* only is _____.

4. Find the number of points where $f(x)=[\sin x+\cos x]$ (where [.] denotes greatest integral function), $x\in[0,2\pi]$ is not continuous, is/are _____.

5. The order of the differential equation of family of curves $y=C_1\sin^{-1}x+C_2\cos^{-1}x+C_3\tan^{-1}x+C_4\cot^{-1}x$ (where C_1, C_2, C_3 & C_4 are arbitrary constants) is _____.

6. If α & β be the roots of the equation $x^2+3x+1=0$ then the value of $\left(\frac{\alpha}{1+\beta}\right)^2+\left(\frac{\beta}{\alpha+1}\right)^2$ is equal to _____.

7. Consider $S_n=\frac{8}{5}+\frac{16}{65}+.....+\frac{8n}{4n^4+1}$. Sum of infinite terms of this series will be _____.

8. If the ratio of the co-efficient to x^{15} to the term independent of x in the expansion of $\left(x^2+\frac{2}{x}\right)^{15}$ is $m:n$, where m& n are coprime, then find the value of $m+n$.

9. The value of the definite integral $\int_{19}^{37}\left(\{x\}^2+3(\sin 2\pi x)\right)dx$ where $\{x\}$ denotes the fractional part function is _____.

10. Let a function f be defined as $f:\{1,2,3,4\}\to\{1,2,3,4\}$. If f satisfy $f(f(x))=f(x)$, $\forall\ x\in\{1,2,3,4\}$, the number of such functions is _____..

SPACE FOR ROUGH WORK

End of 2024 | Mock JEE Main - 8 (CBT) |

Solutions to Mock JEE MAIN – 8 (CBT) | JEE 2024

PHYSICS

SECTION-1

1.(A) In case of a single slit diffraction pattern as shown below, angular position of nth minima is given by
Angular position of nth minima is given by

$$\sin\theta_n = \frac{n\lambda}{D} \qquad \ldots(i)$$

Given $\theta_2 = 60^\circ$ & $\lambda = 6000\times10^{-8}\,cm$; $\sin\theta_2 = \frac{2\lambda}{D}$

We have, $\sin60^\circ = \frac{2\lambda}{D}$; Now, $\sin\theta_1 = \frac{\lambda}{D} = \frac{\sin60^\circ}{2} = 0.43 = \sin25^\circ$

2.(D) The equation of electric field occurring in Y-direction

$$E_y = 66\cos2\pi\times10^{11}\left(t-\frac{x}{c}\right)$$

Therefore, for the magnetic field in Z-direction

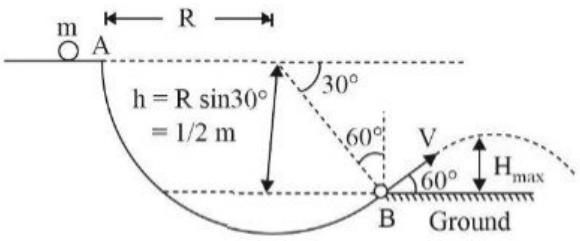
$$B_z = \frac{E_y}{c} = \left(\frac{66}{3\times10^8}\right)\cos2\pi\times10^{11}\left(t-\frac{x}{c}\right)$$

$$= 22\times10^{-8}\cos2\pi\times10^{11}\left(t-\frac{x}{c}\right) = 2.2\times10^{-7}\cos2\pi\times10^{11}\left(t-\frac{x}{c}\right)$$

3.(B) $v = \sqrt{2gh}$

$$= \sqrt{2\times10\times\frac{1}{2}} = \sqrt{10}\,m/s$$

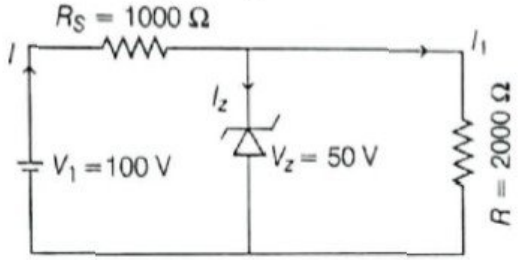
$$H_{max} = \frac{v^2\sin^2 60^\circ}{2g} = \frac{10\times3}{4\times2\times10} = \frac{3}{8}m$$



4.(C) $V_L = 8V,\ V_R = 6V,\ V = \sqrt{V_L^2 + V_R^2} = 10V$

Power factor $= \cos\phi = \frac{V_R}{V} = \frac{6}{10} = 0.6$

5.(A) Consider the given figure and draw the direction of current as follows



Here current I is flowing through resistance R_S & I_1 is the current flowing through $R = 2000\Omega$

$\therefore$ Current $I = \frac{V_1 - V_2}{R_S} \Rightarrow I = \frac{100-50}{1000} = 50mA$ … (i)

[$\because$ Resistance and Zener diode are in parallel, therefore voltage across them are same]

Current $I_1 = \dfrac{50}{2000} = 25mA$ … (ii)

By using Kirchhoff's current law,

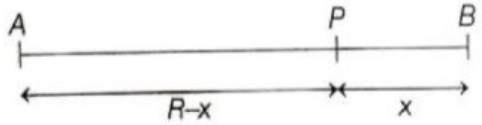
$\because \quad I = I_1 + I_z$

$\Rightarrow \quad I_z = I - I_1$ … (iii)

From equation (i) and (ii) we get

$\Rightarrow \quad I_z = (50-25)\,mA; \qquad I_z = 25mA$

6.(A) Net potential is zero at the point P between A and B



Then, $\dfrac{4qa}{(R-x)^2} = \dfrac{2qa}{(x)^2}; \qquad 2x^2 = (R-x)^2$

$\Rightarrow \quad \sqrt{2}x = R - x \;\Rightarrow\; x = \dfrac{R}{\sqrt{2}+1}$

Distance from A is $\Rightarrow \quad R - x = R - \dfrac{R}{\sqrt{2}+1} = \dfrac{\sqrt{2}R}{\sqrt{2}+1}$

7.(C) For monoatomic gas, $\gamma = 1 + \dfrac{2}{f} = 1 + \dfrac{2}{3} = \dfrac{5}{3}$

In adiabatic condition, $PV^{\gamma} = \text{constant}$

$\Rightarrow \quad P \cdot \gamma V^{\gamma-1} dV + (dP)V^{\gamma} = 0 \quad \Rightarrow \quad \gamma P \dfrac{dV}{V} + (dP) = 0$

$\Rightarrow \quad -\dfrac{dP}{\left(\dfrac{dV}{V}\right)} = \gamma_P \;\Rightarrow\; \beta = \dfrac{5}{3}P$

8.(A) Apply string constant

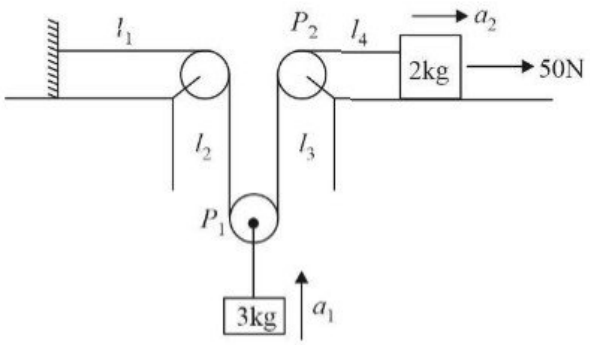
$l_1 + l_2 + l_3 + l_4 = \text{constant}$

$\Rightarrow \quad \dfrac{dl_1}{dt} + \dfrac{dl_2}{dt} + \dfrac{dl_3}{dt} + \dfrac{dl_4}{dt} = 0$

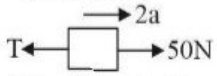
$\Rightarrow \quad 0 + (0 - a_1) + (0 - a_1) + a_2 = 0$

$a_2 = 2a_1$

Let $a_1 = a,\ a_2 = 2a$



FBD of 2kg



$50 - T = 2 \times 2a = 4a$ … (i)

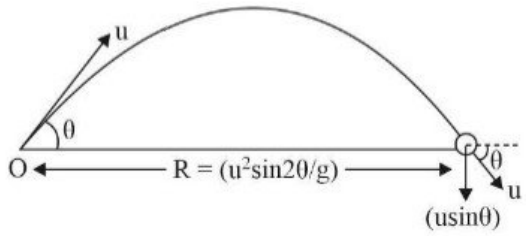
$2T - 30 = 3a$ … (ii)

Equation 1 $\times$ 2 + equation (ii)

$70 = 11a; \qquad a = \dfrac{70}{11} m/s^2$

9.(C) $L_0 = m(u\sin\theta)R = 2(mu\sin\theta)\dfrac{u^2\sin\theta\cos\theta}{g}$

$= \left(\dfrac{2mu^3}{g}\right)\sin^2\theta\cos\theta$



10.(B) Any polytropic process

$PV^N = \text{constant}$

$\Rightarrow \quad \dfrac{nRT}{V}V^N = \text{constant} \quad \Rightarrow \quad TV^{N-1} = \text{constant}$

Given condition $TV^{3/2} = \text{constant}$

$N-1 = \dfrac{3}{2}; \qquad N = \dfrac{5}{2}$

$C = C_V + \dfrac{R}{1-N}; \qquad C = \dfrac{f\times R}{2} + \dfrac{R}{1-N} = \dfrac{3R}{2} + \dfrac{R}{1-\frac{5}{2}} = \left(\dfrac{3}{2} - \dfrac{2}{3}\right)R; \qquad C = \dfrac{5}{6}R$

11.(A) $V_{rms} = \sqrt{\dfrac{V_1^2 + V_2^2 + \ldots\ldots + V_{100}^2}{100}}$

$= \sqrt{\dfrac{\sum_{i=1}^{100} V_i^2}{100}} = \sqrt{\dfrac{(2\times1)^2 + (2\times2)^2 + (2\times3)^2 + \ldots\ldots + (2\times100)^2}{100}}$

$= \dfrac{2}{10}\sqrt{1^2 + 2^2 + \ldots\ldots + 100^2} \qquad = \dfrac{2}{10}\sqrt{\dfrac{100(100+1)(200+1)}{6}}$

$= \dfrac{2}{10}\times10\times\sqrt{\dfrac{101\times201}{6}} = \sqrt{\dfrac{2}{3}\times101\times201} = \sqrt{13534}\,m/s$

12.(B) **I.** $Y = \dfrac{(F/A)}{(\Delta l/l)} = kg\dfrac{m}{S^2}\cdot m^{-2} = \left[ML^{-1}T^{-2}\right]$

II. $F = \eta A\left(\dfrac{dV}{dx}\right) \Rightarrow \left[\eta = \dfrac{F\cdot dx}{AdV} = kg\dfrac{m}{S^2}\cdot\dfrac{m}{m^2}\cdot\dfrac{S}{m}\right] \equiv \left[ML^{-1}T^{-1}\right]$

III. $F = \dfrac{1}{4\pi\varepsilon_0}\cdot\dfrac{q_1q_2}{r^2} \Rightarrow \varepsilon_0 = [M^{-1}L^{-3}T^4A^2]$

IV. Unit of terminal velocity is m/s $[M^0LT^{-1}]$

13.(A) When S is open

Ammeter reading = $I = \dfrac{V}{2R}$ & Voltmeter reading = $IR = \dfrac{V}{2}$

When S is closed

$I' = \dfrac{V}{R}$ and Voltmeter reading $= I'R = V$

Hence, reading of voltmeter and ammeter increases.

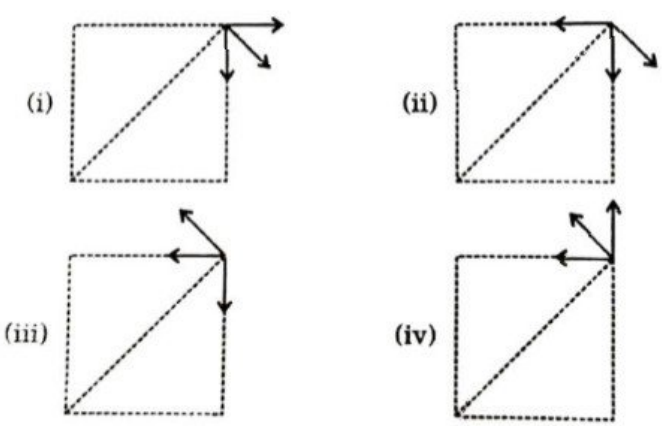
14.(B) $P = VI$

For secondary: $V_2 = \frac{P_2}{I_2} = \frac{500}{12.5} = 40\ volt$

For an ideal transformer (100% efficient)

$P_{input} = P_{output} \quad \Rightarrow \quad V_1 I_1 = V_2 I_2 \quad \Rightarrow \quad I_1 = \frac{V_2 I_2}{V_1} = \frac{40(12.5)}{40 \times 5} = 2.5\ A$

15.(A)



$B_I = B_{IV} > B_{III} = B_{II}$

16.(B) $KE_{max} = eV_0$ [V_0 = stopping potential]

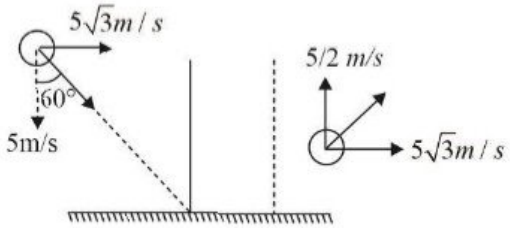
$\Rightarrow \quad 4eV = eV_0 \quad \Rightarrow \quad V_0 = 4volt$

17.(A) $V = \frac{-GM}{(R+h)}, \quad g = \frac{-GM}{(R+h)^2}; \quad \frac{g}{V} = \frac{1}{(R+h)} \quad \Rightarrow \quad h = \frac{V}{g} - R$

18.(B) For metallic conductor as $T \uparrow \Rightarrow R \uparrow$

Slope of I-V curve $= \frac{1}{R} \propto \frac{1}{T}; \quad T_1 < T_2$

19.(B)



$I = \Delta P = m(V_f - V_i)$ along common normal

$= 1 \times \left(\frac{5}{2} - (-5)\right) = \frac{15}{2} = 7.5 N - s$

20.(A) $E_n = \frac{1}{2} m v^2 \quad \ldots$ (i)

$mvr = \frac{nh}{2\pi} \quad \ldots$ (ii)

Equation (i) divided by (ii)

$\frac{\omega}{2} = \frac{2\pi E_0}{nh} \quad \Rightarrow \quad f = \frac{\omega}{2\pi} = \frac{2E_n}{nh}$

SECTION-2

1.(27) Capacitance of charged capacitor, $C_1 = 4\mu F = 4\times10^{-6}F$

Supply voltage, $V_1 = 200V$

Electrostatic energy stored in C_1 is given by

$$E_1 = \frac{1}{2}C_1V_1^2 \quad = \frac{1}{2}\times4\times10^{-6}\times(200)^2 = 8\times10^{-2}J$$

Capacitor of an uncharged capacitor, $C_2 = 2\mu F = 2\times10^{-6}F$

When, C_2 is connected to the circuit the potential acquired by it is V_2

According to the conservation of charge, initial charge on capacitor C_1 is equal to the final charge on capacitor, C_2 and C_2

$$\therefore \qquad V_2(C_1+C_2) = C_1V_1; \quad V_2\times(4+2)\times10^{-6} = 4\times10^{-6}\times200$$

$$V_2 = \frac{400}{3}V$$

Electrostatic energy for the combination of two capacitors is given by

$$E_2 = \frac{1}{2}(C_1+C_2)V_2^2 \quad = \frac{1}{2}(2+4)\times10^{-6}\times\left(\frac{400}{3}\right)^2 \quad = 5.33\times10^{-2}J$$

Hence, amount of electrostatic energy lost by capacitor $C_1 = E_1 - E_2$

$$= 0.08 - 0.0533 = 0.0267 \qquad = 2.67\times10^{-2}J = 26.7mJ \approx 27mJ$$

2.(1) $v = 0 = 9 - 2t$

$t = 4.5\text{sec}$

$$\left[S_5 = 2\times\left[\frac{1}{2}\times2\times\left(\frac{1}{2}\right)^2\right] = \frac{1}{2}m\right]$$

$2S_5 = 1m$

3.(6) If electron comes down from n^{th} state to m^{th} state

then number of lines $= \dfrac{(n-m)(n-m+1)}{2}$

Here n^{th} excited state to 2nd excited state

$n = n+1 \qquad m = 3$

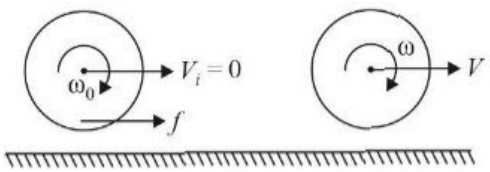
Number of lines $= \dfrac{(n-2)(n-1)}{2} = 10$

$(n-2)(n-1) = 20; \qquad n = 6$

4.(3) Apply conservation of angular momentum about point of contact

$(L_0)_i = (L_0)_f$

$$\Rightarrow \qquad \left(\frac{mR^2}{2}\right)\omega_0 = \frac{mR^2}{2}\omega + (mVR)$$



$$\frac{\omega_0}{2} = \frac{\omega}{2} + \frac{V}{R} = \frac{\omega}{2} + \omega = \frac{3}{2}\omega \quad \Rightarrow \quad \frac{\omega_0}{\omega} = 3$$

5.(11) $f_0 =$ fundamental frequency $= \left(\frac{V}{4l}\right)$

$f_1 = 40\,Hz$ (Here V $\rightarrow$ speed of sound in air)

$f_2 = 60\,Hz$

$f_1 = \frac{V}{4l_1}$; $l_1 = \frac{V}{4f_1}$ & $l_2 = \frac{V}{4f_2}$

Mass of water poured $= \delta \times$ volume

$= 10^3 \times (16\times10^{-4})[l_1 - l_2]$ $= \frac{16}{10}\times\left(\frac{V}{4f_1} - \frac{V}{4f_2}\right)$

$= \frac{4V}{10}\left(\frac{1}{f_1} - \frac{1}{f_2}\right) = \frac{V}{10}\left(\frac{1}{40} - \frac{1}{60}\right)$ $= \frac{4V}{10}\times\frac{20}{40\times60} = \frac{330\times20\times4}{10\times40\times60} = \frac{3.3}{30} = 1.1kg$

$x = 1.1kg$

6.(15) Flux linked with the coil is ; $\phi = B.A. = BA\cos\omega t$

Magnitude of emf induced in coil ; $= \left|\frac{d}{dt}\phi\right| = |BA\omega\sin\omega t|$

Maximum value of induced emf $= BA\omega = \frac{BA2\pi}{T}$

$= \frac{3\times10^{-5}\times\pi\times(0.1)^2\times2\pi}{0.4}$ $= 15\times10^{-6} = 15\mu V$

7.(5) $E_{net} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{\frac{6}{1} + \frac{2}{3}}{\frac{1}{1} + \frac{1}{3}} \approx \frac{20}{4} = 5V$

8.(15) Let v be the position of image, h_i be the height of image and h_0 be the height of object.

Given, $h_i = h_0$

$m_1 = -m_2$; $\frac{f}{u_1 + f} = -\frac{f}{u_2 + f}$ $\therefore$ $-10 + f = -(-20 + f)$

$\Rightarrow$ $-10 + f = 20 - f$; $30 = 2f$; $f = 15cm$

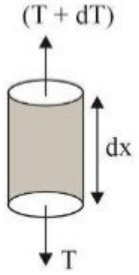
9.(6) $T = \left(\frac{mg}{l}\right)x$

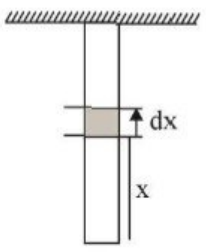
$dU =$ potential energy stored in dx length

$dU = \frac{1}{2}\times(\sigma\times\varepsilon)\times$ volume

$= \frac{1}{2}\times\left(\frac{\sigma^2}{Y}\right)\times$ volume $= \frac{T^2}{2YA^2}\times Adx = \frac{\frac{m^2g^2}{l^2}x^2}{2YA^2}Adx$

$\int dU = \left(\frac{m^2g^2}{2Yl^2A}\right)\int_0^l x^2dx \Rightarrow U = \frac{m^2g^2l^3}{6Yl^2A} = \frac{(m^2g^2l)}{6YA}$





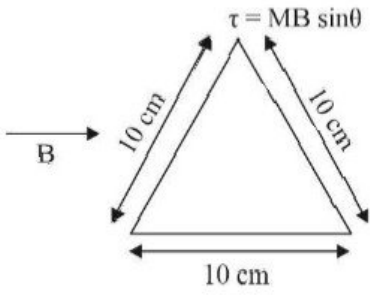
Side of equilateral triangular coil,

$a = 10cm = 0.1m$

Horizontal magnetic field,

$B = 20mT = 20 \times 10^{-3} T$

Current in coil, $i = 0.2A$



Torque acting on coil, $\tau = \sqrt{x} \times 10^{-5} N - m$

Torque acting on coil can be calculated as

Here, M is pole strength and B is magnetic field.

Magnetic pole strength, $M = iA$

$$\tau = i\left(\frac{\sqrt{3}}{4}a^2\right)B\sin\theta \qquad [\because A = \frac{\sqrt{3}}{4}a^2 \text{ (for equilateral triangle)}]$$

$$= 0.2 \times \frac{\sqrt{3}}{4} \times (0.1)^2 \times 20 \times 10^{-3} \times \sin 90^\circ \qquad [\because A \perp B]$$

$$= \sqrt{3} \times 10^{-5} N - m$$

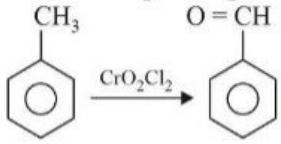
After comparing obtained value with the given value $x = 3$.

CHEMISTRY

SECTION-1

1.(D) Substitution is very difficult in aryl halide.

2.(C) Chromyl chloride oxidizes toluene to benzaldehyde. This reaction is known as the Etard reaction. The solvent used is carbon disulfide. Chromyl chloride converts the methyl group to a chromium complex. This complex on acid hydrolysis gives the corresponding aldehyde.



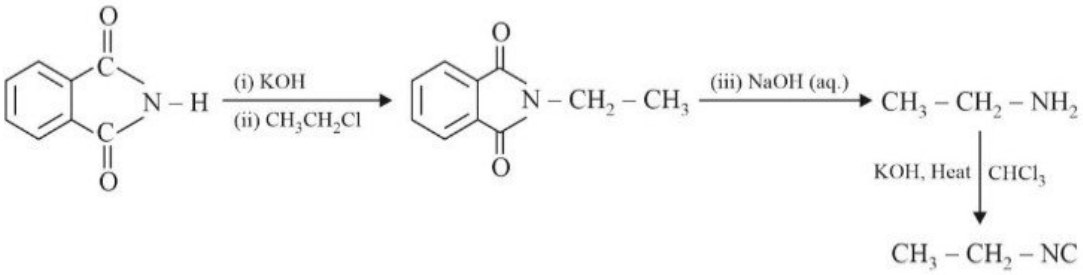
3.(C) **I.** $cis-[Co(NH_3)_4Cl_2]^+$ has plane of symmetry as well as centre of symmetry, so optically inactive.

II. trans-$[Co(en)_2Cl_2]^+$ has plane of symmetry as well as centre of symmetry, so optically inactive.

III. $cis-[Co(en)_2Cl_2]^+$ lacks plane of symmetry as well as centre of symmetry, so optically active.

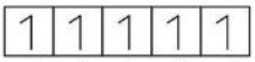
IV. $[Co(en)_3]^{3+}$ lacks plane of symmetry, so optically active.

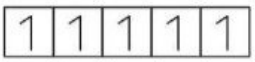
4.(D)



5.(D) Sulphate is estimated as $BaSO_4$ not as $MgSO_4$ because $MgSO_4$ is soluble in water and does not form a precipitate.

The ionic radius of Mg^{2+} is smaller than that of Ba^{2+} because of moving down in a group, ionic radius increases.

6.(B) $Mn^{2+} \rightarrow d^5 \rightarrow$ 

$Fe^{3+} \rightarrow d^5 \rightarrow$ 

$Ni^{2+} \rightarrow d^8 \rightarrow$ 

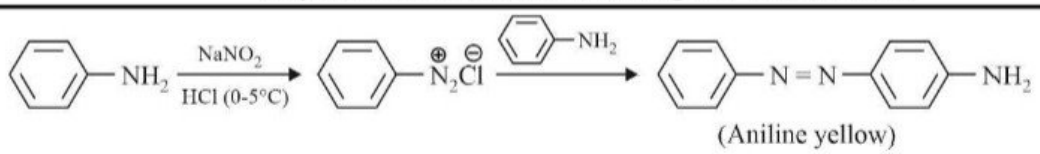
7.(B) Refer VSEPR theory.

8.(C) C is correct IUPAC name.

9.(A) Be has completely filled stable $2s^2$ orbital and thus Be has higher ionization energy than B. 2s orbital has less energy then 2p orbital. (From $(n+\ell)$ rule).

10.(C) Fact based

11.(B) Valence shell electronic configuration of Eu is $4f^7 6s^2$.

12.(D) 

(Aniline yellow)

13.(D) Carbocation is stabilized by lone pair of 'O' (mesomeric stabilized carbocation).

14.(B) Anti-Markovnikov's addition takes place in hydroboration reaction.

15.(A) For single e^- species
Energy α n

16.(A) Ony liquid freezes at freezing point. Thus equilibrium between solid and liquid forms exist at freezing point.

17.(A) $Co^{2+}\left(d^7\right)$

Since SCN^- acts like a weak field ligand so no pairing of e^-s occur in Co^{2+} ion. Complex formed will have sp^3 hybridization and tetrahedral geometry.

18.(D) Kjeldahl method is not applicable to nitro, azo and compound containing N in ring.

19.(B) [X] : Rosenmund reaction
[Y] : Stephen's reaction
[Z] : Reduction of ester using $DIBA\ell - H$

20.(B) Amylose gives a blue colour with iodine solution due to formation of an inclusion complex in starch

SECTION-2

1.(10) Electronic configuration of O_2^+

$\sigma_{1s^2}\ \overset{*}{\sigma}_{1s^2}\ \sigma_{2s^2}\ \overset{*}{\sigma}_{2s^2}\ \sigma^2_{2p_z}\ \pi_{2p_x^2}\pi_{2p_y^2}\ \overset{*}{\pi}_{2p_x^1}$

No. of e^-s in BMO : 10

2.(9) $K_b = 8\times10^{-10}$

$C_5H_5NH^+Cl^-$ it is a salt of strong acid and weak base

$$\therefore \quad pH = 7-\frac{1}{2}pK_h-\frac{1}{2}\log C \quad \Rightarrow \quad pH = 7-\frac{1}{2}pK_b-\frac{1}{2}\log 0.25$$

$$\Rightarrow \quad pK_b = 9.1 \quad \Rightarrow \quad x = 9$$

3.(8) Balance the equation by any method

$$4Zn+10HNO_3 \longrightarrow 4Zn(NO_3)_2+3H_2O+NH_4NO_3 \qquad \therefore \qquad a+b+c=4+3+1=8$$

4.(40) $\frac{1}{2}N_2(g)+\frac{3}{2}H_2(g) \longrightarrow NH_3(g)$

$$\Delta S_{rxn} = 50-\frac{1}{2}\times60-\frac{3}{2}\times xJK^{-1}; \qquad \Delta H_{rxn} = -30000J$$

At equilibrium; $\Delta G_{rxn} = 0 = \Delta H_{rxn} - T\Delta S_{rxn}$

$$750\times\left(20-\frac{3}{2}x\right) = -30000; \quad x = \frac{120}{3} = 40$$

5.(7) Benzophenone, 3-Pentanone, $(CH_3CH_2)_2CH-OH$, CH_3COOH, CH_3CONH_2, $CH_3-COOCH_3$, CH_3-COCl will not give iodoform test. Group $\left(\begin{matrix}CH_3-C-\\ \\ \\ O\end{matrix}\right)$ of carbonyl compounds gives positive iodoform test but acids and acid derivatives do not.

6.(34) $\%Br = \frac{80}{188} \times \frac{0.12}{0.15} \times 100 = 34$

7.(15) Atomic no. = 115
Group = 15

8.(2) For 99% completion, $t_{99\%} = \frac{2.303}{k} \times \log \frac{100}{100-99}$ ...(i)

For 90% completion, $t_{90\%} = \frac{2.303}{k} \times \log \frac{100}{100-90}$ ...(ii)

Using equations (i) and (ii) $\frac{t_{99\%}}{t_{90\%}} = \frac{\log 100}{\log 10} = 2$

9.(42) Let x be the mass of $CaCO_3$ hence mass of $MgCO_3 = 92 - x$

$$CaCO_3 \longrightarrow CaO + CO_2$$
$$\left(\frac{x}{100}\right) \qquad \left(\frac{x}{100}\right)$$
$$MgCO_3 \longrightarrow MgO + CO_2$$
$$\left(\frac{92-x}{84}\right) \qquad \left(\frac{92-x}{84}\right)$$

Mass of residue = 48 g

$$\Rightarrow \quad \frac{x}{100} \times 56 + \frac{92-x}{84} \times 40 = 48; \quad x = 50g$$

$$\therefore \quad \text{Mass of } MgCO_3 = 92 - 50 = 42g.$$

10.(18) Molarity $= \frac{10 \times 1.8 \times 98}{98} = 18M$

MATHEMATICS

SECTION-1

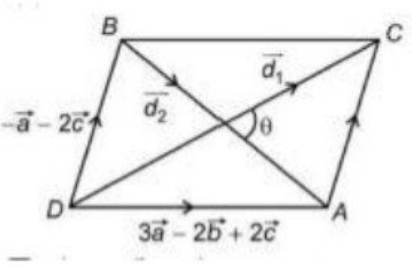
1.(A) $\vec{d}_1 = (3\vec{a} - 2\vec{b} + 2\vec{c}) + (-\vec{a} - 2\vec{c})$

$= 2\vec{a} - 2\vec{b}$

$(-\vec{a} - 2\vec{c}) + \vec{d}_2 = 3\vec{a} - 2\vec{b} + 2\vec{c}$

$\Rightarrow \quad \vec{d}_2 = (3\vec{a} - 2\vec{b} + 2\vec{c}) - (-\vec{a} - 2\vec{c})$

$= 4\vec{a} - 2\vec{b} + 4\vec{c}$



Let θ be the angle between given two diagonals

Then $\cos\theta = \dfrac{\vec{d}_1 \cdot \vec{d}_2}{|\vec{d}_1||\vec{d}_2|} = \dfrac{8+4}{\sqrt{8}\sqrt{16+4+16}}$

$= \dfrac{12}{2\sqrt{2}\cdot 6} = \dfrac{1}{\sqrt{2}} \quad \Rightarrow \quad \theta = \cos^{-1}\dfrac{1}{\sqrt{2}} = \dfrac{\pi}{4}$

2.(D) $x^2 + (2 - \tan\theta)x - (1 + \tan\theta) = 0$

$\alpha + \beta = \tan\theta - 2; \qquad \alpha\beta = -\tan\theta - 1$

From equation (i) and (ii)

$\alpha + \beta + \alpha\beta = -3; \qquad (\alpha + 1)(\beta + 1) = -2$

Hence either $\alpha + 1 = -2$ & $\beta + 1 = 1$

Then $\alpha = -3, \beta = 0$

OR $\quad \alpha + 1 = -1$ & $\beta + 1 = 2; \qquad \alpha = -2$ & $\beta = 1$

If $\alpha = -3, \beta = 0$ then $\tan\theta = -1 \quad \Rightarrow \quad \alpha = \dfrac{3\pi}{4}, \dfrac{7\pi}{4}$

If $\alpha = -2$ & $\beta = 1$ then $\tan\theta = 1 \quad \Rightarrow \quad \theta = \dfrac{\pi}{4}, \dfrac{5\pi}{4}$

Sum $\dfrac{3\pi}{4} + \dfrac{7\pi}{4} + \dfrac{\pi}{4} + \dfrac{5\pi}{4} = 4\pi \quad \Rightarrow \quad k = 4$

3.(D) $D_1 = \begin{vmatrix} 4 & 1 & 1 \\ 6 & 1 & 3 \\ q & 2 & p \end{vmatrix} = -2(p - q + 6),\ D_2 = \begin{vmatrix} 1 & 4 & 1 \\ 2 & 6 & 3 \\ 1 & q & p \end{vmatrix} = -2p - q + 6,\ D_3 = \begin{vmatrix} 1 & 1 & 4 \\ 2 & 1 & 6 \\ 1 & 2 & q \end{vmatrix} = -q + 6$

$D = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 1 & 2 & p \end{vmatrix} = -p; \qquad x = \dfrac{D_1}{D} = \dfrac{2(p - q + 6)}{p},\ y = \dfrac{D_2}{D} = \dfrac{2p + q - 6}{p},\ z = \dfrac{D_3}{D} = \dfrac{q - 6}{p}$

Now for unique solution $p \neq 0,\ q \in R$

$\Rightarrow$ total no. of order pairs $(p, q) = 20 \times 21 = 420$

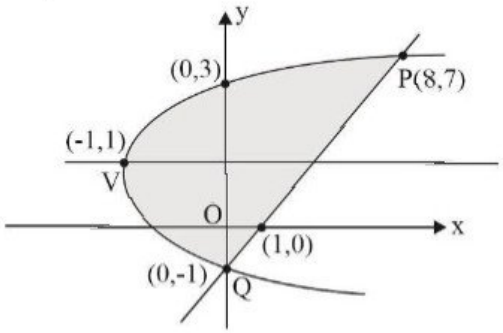
For no solution $p = 0,\ q \neq 6$

$\Rightarrow$ total no. of ordered pairs of $(p, q) = 1 \times 20 = 20$

For infinite many solution $p = 0,\ q = 6$

$\Rightarrow$ total no of ordered pair $(p, q) = 1 \times 1 = 1$

4.(D) $A=\int_{y=-1}^{y=7}(y+1)-\left[-1+\frac{(y-1)^2}{4}\right]=\frac{64}{3}$



5.(B)

Class	Frequency f_i	Cumulative frequency (cf)
95 – 105	9	9
105 – 115	13	22
115 – 125	25	47
125 – 135	30	77
135 – 145	13	90
145 – 155	10	100
	$N=\Sigma f_i=100$	

Thus $N=100$ and therefore $\frac{N}{2}=50$

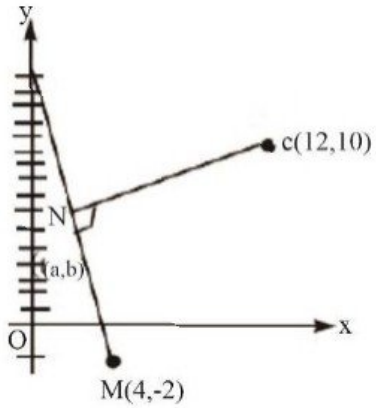
$\Rightarrow$ median class is 125 – 135

$\Rightarrow$ $L=125,\ f=30,\ h=10\ \&\ c=47$

$\therefore$ $\text{median}=L+\frac{\left(\frac{N}{2}-c\right)}{f}\times h$

$=\left\{125+\frac{(50-47)}{30}\times 10\right\}=(125+1)=126$

6.(B) (a, b) is foot of the perpendicular from (12, 10) on the line $y+5x=18$



$\Rightarrow$ $\frac{a-12}{5}=\frac{b-10}{1}=-\left(\frac{10+5(12)-18}{26}\right)=-2$ $\Rightarrow$ $(a,b)=(2,8)$

7.(B) $[2x-3]=-1,0,1 \Rightarrow -1\le 2x-3<2 \Rightarrow 1\le x<\frac{5}{2}$

$$\begin{cases}4-x>0\\4-x\neq 1\\x-1>0\end{cases} \Rightarrow x\in(1,3)\cup(3,4)$$

Intersection of both is $x\in\left(1,\frac{5}{2}\right)$

8.(D) $a_1+a_4+a_7+.....+a_{16}=\frac{6}{2}(2a+5(3d))=147 \Rightarrow 2a+15d=49$

Using $AM\ge GM$

$$\frac{\sum a_i}{16}\ge(a_1a_2.....a_{16})^{1/16}$$

$$\therefore \quad a_1a_2....a_{16}\le\left(\frac{a_1+a_2+.....+a_{16}}{16}\right)^{16}\le\left(\frac{2a+15d}{2}\right)^{16}\le\left(\frac{49}{2}\right)^{16}$$

$$\Rightarrow \quad S=49 \ \&\ W=2 \qquad \therefore \qquad S+W=49+2=51$$

9.(C) $\lim\limits_{x\to 0}\dfrac{\int_0^x\frac{t^2dt}{\sqrt{a+t}}}{bx-\sin x}$

Using L' hospital; $\quad L=\lim\limits_{x\to 0}\dfrac{x^2}{(b-\cos x)\sqrt{a+x}}$

For the non-zero existence of limit $b=1$

$$L=\lim_{x\to 0}\frac{x^2}{(1-\cos x)\sqrt{a+x}}=1 \Rightarrow a=4$$

10.(D) $\because$ $f(x)$ is odd function in $[-2,2]$ $\quad\therefore\quad f(-2)=-f(2)$ and $f(0)=0$

By LMVT rule on $[-2,0]$; $\quad\dfrac{f(0)-f(-2)}{0-(-2)}=f'(c)$ *for* $c\in(-2,0)$

$0-f(-2)=2f'(c) \quad\therefore\quad f(2)=2f'(c)$

11.(C) $\Rightarrow \quad \dfrac{\frac{(xdy-ydx)}{x^2}}{1+\cos\left(\frac{y}{x}\right)}=\dfrac{dx}{x^3} \quad\Rightarrow\quad \int\dfrac{d\left(\frac{y}{x}\right)}{1+\cos\left(\frac{y}{x}\right)}=\int\dfrac{dx}{x^3}$

$$\Rightarrow \quad \frac{1}{2}\int\frac{d\left(\frac{y}{x}\right)}{\cos^2\left(\frac{y}{2x}\right)}=\int\frac{dx}{x^3} \quad\Rightarrow\quad \frac{1}{2}\frac{\tan\left(\frac{y}{2x}\right)}{\frac{1}{2}}=\frac{-1}{2x^2}+c$$

$$\Rightarrow \quad \frac{1}{2}\frac{\tan\left(\frac{y}{2x}\right)}{\frac{1}{2}}=\frac{-1}{2x^2}+c \quad\Rightarrow\quad \tan\left(\frac{y}{2x}\right)+\frac{1}{2x^2}=c \quad\Rightarrow\quad p+q=4$$

12.(D) Given: $f(x)=\begin{vmatrix} \sin^2 x & -2+\cos^2 x & \cos 2x \\ 2+\sin^2 x & \cos^2 x & \cos 2x \\ \sin^2 x & \cos^2 x & 1+\cos 2x \end{vmatrix}$

$R_2 \to R_2 - R_1; \qquad R_3 \to R_3 - R_1$

$\Rightarrow \quad f(x)=\begin{vmatrix} \sin^2 x & -2+\cos^2 x & \cos 2x \\ 2 & 2 & 0 \\ 0 & 2 & 1 \end{vmatrix}$

$\Rightarrow \quad f(x)=2\sin^2 x+4-2\cos^2 x+4\cos 2x$

$\Rightarrow \quad f(x)=2\cos 2x+4 \quad \therefore \quad f(x)_{\max}=6$

13.(C) $\because \quad SP+S'P=2a=10 \quad \Rightarrow \quad S'P=7$

$\because \quad \frac{1}{S'P}+\frac{1}{S'P'}=\frac{2a}{b^2} \quad \Rightarrow \quad \frac{1}{S'P'}=\frac{10}{9}-\frac{1}{7}=\frac{61}{63}$

$\therefore \quad 61(S'P')-56=7$

14.(B) P (min is 3/max is 6) $=\frac{P(\text{min is } 3 \cap \text{max is } 6)}{P(\text{max is } 6)}=\frac{{}^2C_1}{{}^5C_2}=\frac{1}{5}$

15.(B) $(3\vec{a}+\vec{b})\cdot(\vec{a}-4\vec{b}) \quad =3|\vec{a}|^2-11\vec{a}\cdot\vec{b}-4|\vec{b}|^2<0$

Therefore, the angle between $3\vec{a}+\vec{b}$ & $\vec{a}-4\vec{b}$ is obtuse

The longer diagonal is given by $\vec{\alpha}=(3\vec{a}+\vec{b})-(\vec{a}-4\vec{b})=2\vec{a}+5\vec{b}$

Now, $|\vec{\alpha}|^2=|2\vec{a}+5\vec{b}|^2=4|\vec{a}|^2+25|\vec{b}|^2+20\vec{a}\cdot\vec{b}=1744; \qquad |\vec{\alpha}|=4\sqrt{109}$

16.(B) $\because \quad |3+i(z-1)|=|i(z-1-3i)|=|i||z-1-3i|$

$=|z-1-3i| \qquad \ldots(i)$

And $|z-1-3i|=|(z-2i)+(-1-i)|\le|z-2i|+|-1-i| \qquad [\because |z_1+z_2|\le|z_1|+|z_2|]$

$\therefore \quad |z-1-3i|\le|z-2i|+\sqrt{2} \quad \Rightarrow \quad |z-1-3i|\le\sqrt{2}+\sqrt{2} \qquad [\because |z-2i|\le\sqrt{2}]$

$\Rightarrow \quad |z-1-3i|\le 2\sqrt{2}; \qquad$ From eq. (i) we get $|3+i(z-1)|\le 2\sqrt{2}$

Hence, the maximum value of $|3-i(z-1)|$ is $2\sqrt{2}$

17.(B) If l,m,n are direction cosines of two lines are such that

$l+m+n=0 \qquad \ldots$ (i)

$l^2+m^2-n^2=0 \qquad \ldots$ (ii)

$\Rightarrow \quad l^2+m^2-(-l-m)^2=0 \quad \Rightarrow \quad 2lm=0 \Rightarrow l=0 \text{ or } m=0$

If $l=0$, then $n=-m \quad \Rightarrow \quad l:m:n=0:1:-1$

and if $m=0$, then $n=-1 \quad \Rightarrow \quad l:m:n=1:0:-1$

$\therefore \quad \cos\theta=\frac{0+0+1}{\sqrt{0+1+1}\sqrt{0+1+1}}=\frac{1}{2} \quad \Rightarrow \quad \theta=\frac{\pi}{3}$

18.(C) Centres are (10, 0) and (–15, 0)

$r_1 = 6\,;\ r_2 = 9;\qquad d = 25$

$r_1 + r_2 < d \quad \Rightarrow \quad$ circles are separated

$PQ = l = \sqrt{d^2 - (r_1 + r_2)^2} = \sqrt{625 - 225} = 20$

19.(D) Solving, $e^x = e^{a-x}$, we get $e^{2x} = e^a \Rightarrow x = \dfrac{a}{2}$

$$S = \int_0^{a/2} (e^a \cdot e^{-x} - e^x)dx = [-(e^a \cdot e^{-x} + e^x)]_0^{a/2}$$

$$= (e^a + 1) - (e^{a/2} + e^{a/2}) = e^a - 2e^{a/2} + 1 = (e^{a/2} - 1)^2$$

$$\therefore \quad \frac{S}{a^2} = \left(\frac{e^{a/2} - 1}{a}\right)^2 = \frac{1}{4}\left(\frac{e^{a/2} - 1}{a/2}\right)^2 \qquad \therefore \quad \lim_{a \to 0} \frac{S}{a^2} = \frac{1}{4}$$

20.(B) We have $\lim\limits_{n\to\infty} \sum\limits_{k=1}^{n} \dfrac{n}{n^2\left(a + \dfrac{k}{n}\right)\left(b + \dfrac{k}{n}\right)} = \lim\limits_{n\to\infty} \dfrac{1}{n} \sum\limits_{k=1}^{n} \dfrac{1}{\left(a + \dfrac{k}{n}\right)\left(b + \dfrac{k}{n}\right)}$

$$= \int_0^1 \frac{1}{(a+x)(b+x)}dx = \frac{1}{a-b}\int_0^1 \frac{(a+x)-(b+x)}{(a+x)(b+x)}dx = \frac{1}{a-b}\int_0^1 \left(\frac{1}{b+x} - \frac{1}{a+x}\right)dx$$

$$= \frac{1}{a-b}[\ln(b+x) - \ln(a+x)]_0^1 = \frac{1}{a-b}\left[\ln\frac{(b+x)}{(a+x)}\right]_0^1 = \frac{1}{a-b}\ln\frac{(b+1)a}{(a+1)b}$$

Now if $a = b$ then $l = \lim\limits_{n\to\infty} \sum\limits_{k=1}^{n} \dfrac{1}{n} \dfrac{1}{\left(a + \dfrac{k}{n}\right)^2} = \int_0^1 \frac{dx}{(a+x)^2} = \left.\frac{-1}{a+x}\right|_1^0 = \frac{1}{a} - \frac{1}{a+1} = \frac{1}{a(a+1)}$

SECTION-2

1.(2) Since $\dfrac{e}{2}$ & $\dfrac{e'}{2}$ are the eccentricities of a hyperbola and its conjugate, we have

$$\left(\frac{e}{2}\right)^{-2} + \left(\frac{e'}{2}\right)^{-2} = 1 \text{ or } \frac{4}{e^2} + \frac{4}{(e')^2} = 1$$

$$4 = \frac{e^2 e'^2}{e^2 + e'^2} \qquad \ldots \text{(i)}$$

Equations of variable line is $\dfrac{x}{e} + \dfrac{y}{e'} = 1\,;\ xe' + ye - ee' = 0$

It is tangent to the circle $x^2 + y^2 = r^2$

$$\frac{|0 - ee'|}{\sqrt{e^2 + e'^2}} = r\,; \qquad r = \frac{ee'}{\sqrt{(e^2 + e'^2)}} = 2$$

2.(3) Given lines are $r = 3\hat{i} + 8\hat{j} + 3\hat{k} + \lambda(3\hat{i} - \hat{j} + \hat{k})$

$r = (-3\hat{i} - 7\hat{j} + 6\hat{k}) + \mu(-3\hat{i} + 2\hat{j} + 4\hat{k})$

where λ, μ are parameters

$$\text{Shortest distance} = \left|(\vec{a}_2 - \vec{a}_1) \cdot \frac{(\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|}\right| = \sqrt{270} = 3\sqrt{30}$$

3.(3300) $n(A) = 40\%$ of 10000 = 4000

$n(B) = 20\%$ of 10000 = 2000

$n(C) = 10\%$ of 10000 = 1000

$n(A \cap B) = 5\%$ of 10000 = 500

$n(B \cap C) = 3\%$ of 10000 = 300

$n(C \cap A) = 4\%$ of 10000 = 400

$n(A \cap B \cap C) = 2\%$ of 10000 = 200

We want to find $n(A \cap B^c \cap C^c) = n[A \cap (B \cup C)^c]$

$= n(A) - n[A \cap (B \cup C)] = n(A) - n[(A \cap B) \cup (A \cap C)]$

$= n(A) - [n(A \cap B) + n(A \cap C) - n(A \cap B \cap C)]$

$= 4000 - [500 + 400 - 200] = 4000 - 700 = 3300$

4.(5) We know [.] is not continuous at integral points

Thus, $f(x) = [\sin x + \cos x]$ will be discontinuous at those points, where $\sin x + \cos x$ is an integer,

i.e., $x \in \left\{\frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{3\pi}{2}, \frac{7\pi}{4}\right\}$

Thus, the number of points at which $f(x)$ is discontinuous is 5.

5.(3) Here, $y = C_1 \sin^{-1} x + C_2 \cos^{-1} x + C_3 \tan^{-1} x + C_4 \cot^{-1} x$

$$\Rightarrow \quad y = C_1 \sin^{-1} x + C_2\left(\frac{\pi}{2} - \sin^{-1} x\right) + C_3 \tan^{-1} x + C_4\left(\frac{\pi}{2} - \tan^{-1} x\right)$$

$$= (C_1 - C_2)\sin^{-1} x + (C_3 - C_4)\tan^{-1} x + (C_2 + C_4)\frac{\pi}{2}$$

Since there are three arbitrary constant. Hence, order of the differential equation is 3.

6.(18) $\alpha + \beta = -3$; $\alpha\beta = 1$, also $\alpha^2 + 3\alpha + 1 = 0$ and $\beta^2 + 3\beta + 1 = 0$

where $\alpha^2 = -(3\alpha + 1)$ & $\beta^2 = -(3\beta + 1)$; $\qquad E = \frac{\alpha^2}{(1+\beta)^2} + \frac{\beta^2}{(\alpha+1)^2}$

$$E = \frac{\alpha^2}{1 + 2\beta + \beta^2} + \frac{\beta^2}{1 + 2\alpha + \alpha^2} = \left(\frac{-(3\alpha + 1)}{-\beta}\right) + \left(\frac{-(1 + 3\beta)}{-\alpha}\right)$$

$$y = \frac{1 + 3\alpha}{\beta} + \frac{1 + 3\beta}{\alpha} = \frac{\alpha(1 + 3\alpha) + \beta(1 + 3\beta)}{\alpha\beta} \quad (\text{as } \alpha\beta = 1)$$

$$= 3(\alpha^2 + \beta^2) + (\alpha + \beta) = 3[9 - 2] + (-3) = 21 - 3 = 18$$

7.(2) $T_r = \dfrac{8r}{4r^4+1} = \dfrac{8r}{(4r^4+4r^2+1)-4r^2} = \dfrac{8r}{(2r^2+1)^2-(2r)^2}$

$= \dfrac{8r}{(2r^2-2r+1)(2r^2+2r+1)} = 2\left[\dfrac{1}{(2r^2-2r+1)} - \dfrac{1}{(2r^2+2r+1)}\right]$

$S_n = 2\left[1-\dfrac{1}{2n^2+2n+1}\right] \Rightarrow S_\infty = 2$

8.(33) General term is ${}^{15}C_r\, x^{2r}\left(\dfrac{2}{x}\right)^{15-r}$

$= {}^{15}C_r 2^{15-r} \cdot x^{2r+r-15}$

$= {}^{15}C_r 2^{15-r} \cdot x^{3r-15}$

For coefficient of x^{15}, $3r-15=15,\ r=10$

$\therefore$ coefficient $= {}^{15}C_{10}2^5$

For term independent of x,

$3r-15=0; \quad r=5 \qquad \therefore$ coefficient $= {}^{15}C_{10}2^{10}$

$\therefore$ required ratio $= \dfrac{{}^{15}C_{10}2^5}{{}^{15}C_5 2^{10}} = 1:32 \qquad \therefore \quad m+n=33$

9.(6) Put $x = 19+y \Rightarrow dx = dy$

$$I = \int_0^{18}\left(\{19+y\}^2 + 3\sin 2y\right)dy = \int_0^{18}\left(\{y\}^2 + \underbrace{\sin 2\pi y}_{zero}\right)dy = 18\int_0^1 y^2 dy = 6$$

10.(41) Here, $f(f(x)) = f(x)$, let $f(x) = y$

$\Rightarrow \quad f(y) = y$

Case I : Range contains exactly one element. It can be done in ${}^4C_1$ ways.

Say, $f(1)=1$ remaining 3 elements 2, 3, 4 can be mapped only in one way $\Rightarrow$ Total $= {}^4C_1 \cdot 1 = 4$

Case II : Range contains two elements, this can be done in ${}^4C_2$ ways.

Remaining 2 elements i.e., 3 and 4 each can be mapped in 2 ways

$\therefore$ Total $= {}^4C_2 \times 2 \times 2 = 24$

Case III : Range contains exactly 3 elements which can be done in ${}^4C_3 = 4$. Say

$f(1)=1,\ f(2)=2,\ f(3)=3$

Now, remaining element can be mapped only in 3 ways

$\therefore$ Total $= {}^4C_3 \cdot 3 = 12$ ways

Case IV : Range contains all 4 elements $f(1)=1,\ f(2)=2,\ f(3)=3,\ f(4)=4$

Only 1 way

Total $= 4+24+12+1 = 41$


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